

GROWTH RATES OF FIVE REEF-BUILDING CORALS IN THE NORTHEASTERN CARIBBEAN

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ABSTRACT

Growth rates (linear extension) were measured for five species of reef-building corals in three environmental zones: backreef-lagoon, shallow forereef, and base of forereef on a bank barrier reef in the U.S. Virgin Islands during periods of highest and lowest annual water temperatures. Comparison with similar studies elsewhere in the tropical western Atlantic shows some corals (e.g. *Montastrea annularis*) to be consistent in growth rate regardless of temperature, zone or geographic area, while other corals show differing rates of growth dependent on temperature (*Acropora palmata*, *A. prolifera*), reef zone (*A. palmata*) or geographic area (*A. cervicornis*). The range of estimated annual rates of linear growth for the five corals considered was 6.6–8.9 mm/yr for *M. annularis*, 3.0–3.5 mm/yr for *Porites astreoides*, 71 mm/yr for *A. cervicornis*, 59–82 mm/yr for *A. prolifera*, and 47–99 mm/yr for *A. palmata*. Calcification rate (i.e. the amount of calcium carbonate deposited per unit branch perimeter) did not differ due to temperature, but did show differences due to environmental zone in *Acropora palmata*, averaging 0.85 g/cm/yr in the backreef, 1.60 g/cm/yr in the shallow forereef and 1.35 g/cm/yr in the deep forereef.

A number of methods have been used to estimate the growth rates of scleractinian corals under natural field conditions in the tropical western Atlantic. The most useful of these have been the following: (1) measurement of actual linear increment (length or radius) during the period of study, measured directly in coral fragments attached to artificial bases (Bak, 1976; Lewis et al., 1968; Shinn, 1966; Hoffmesiter and Multer, 1964; Vaughn, 1915), or by reference to a marker in the skeleton, such as a driven spike (Shinn, 1976) or an Alizarin Red S dye marker incorporated into the skeleton at the beginning of the experiment (Barnes, 1972; Dustan, 1975); (2) the use of periodic density variations in the skeleton as indicators of annual growth increment (Baker and Weber, 1975; MacIntyre and Smith, 1974; Aller and Dodge, 1974); (3) measurement of the increase in skeletal weight either by periodically lifting corals out of the water, or by the use of an underwater balance (Bak, 1974, 1976; Vaughn, 1916); (4) determination of the rate of incorporation of radioisotopic calcium into the skeleton (Goreau and Goreau, 1959). Recently the

first two of these methods have been most widely applied under field conditions. The advantages and problems of the various methods have been summarized by Bud-demeier and Kinzie (1976).

A comparison of available data for tropical western Atlantic corals revealed a considerable amount of variation in skeletal growth rate depending on geographical location, probably as a function of local environmental conditions. Generally speaking, insufficient information concerning the local environmental situation has been provided to permit reasonable interpretation of these growth differences between study areas.

In the present study the growth rates of five scleractinian species were estimated in the three major environmental zones of a shallow bank barrier reef (Adey, 1975) at Buck Island Reef National Monument, St. Croix, U.S. Virgin Islands during periods of highest and lowest local water temperature. All five of the species studies are important contributors to the reef framework on this particular reef (Gladfelter et al., 1977).

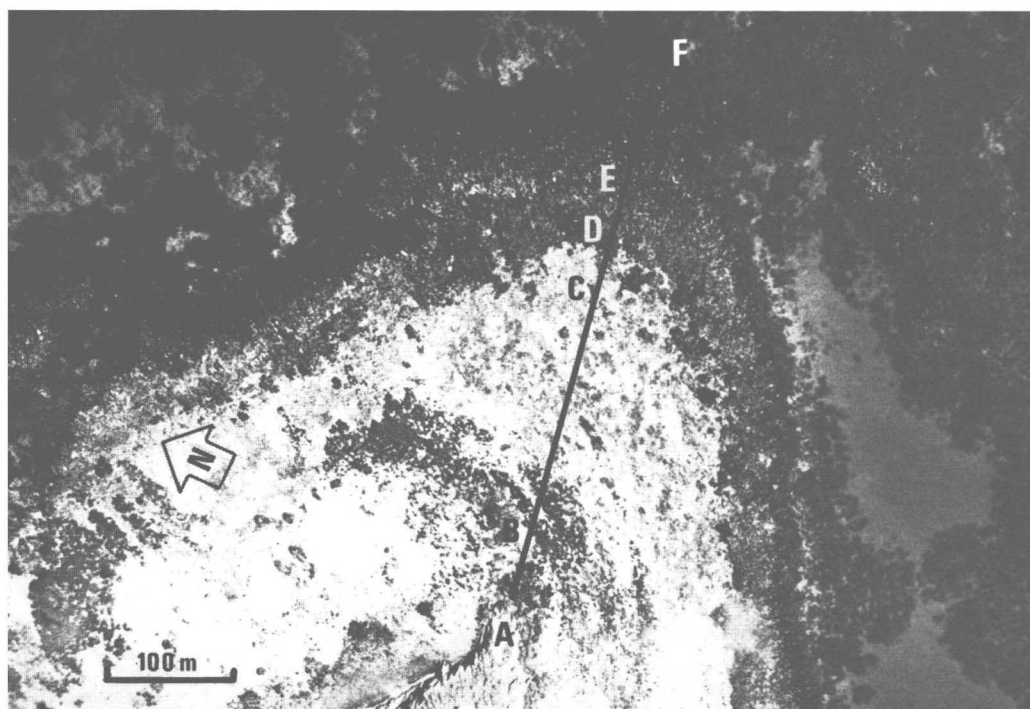


Figure 1. Aerial photograph of the eastern portion of Buck Island (A) and the eastern tip of the bank barrier reef (C-E). The transect (A-F) shows the locations of the corals studied.

METHODS AND MATERIALS

This study was conducted near the eastern tip of the bank barrier reef which nearly encircles Buck Island (Fig. 1). The three major reef zones here are the protected backreef and lagoon (BR), the shallow windward forereef (SFR), and the base of the forereef slope and adjacent bank (DFR). Figure 2 is a schematic cross section through the eastern tip of the reef system, showing the locations of the corals studied. The principal environmental parameters noted in this investigation are presented in Table 1. Temperature was measured 3–5 times per week at a depth of 1.5 m in the lagoon. Most of the coral growth rates were estimated from the results of two staining periods of 2–4 months (during the fall and spring) when the water temperature was at its annual maximum (which remained constant at 29.5°C from July through Novem-

ber) and again when the water temperature was at its annual minimum (which was 26°C from February through April). During each of the staining periods day length and cloud cover were similar. In addition some corals were stained and collected after 8 months of continuous growth.

The five coral species all contribute to the reef framework at Buck Island: (1) *Acropora palmata* dominates the entire eastern portion of the reef from the edge of the backreef to the base of the forereef; (2) *A. prolifera* forms extensive dense thickets within the lagoon; (3) *A. cervicornis* forms thickets on the bank seaward of the base of the south forereef; (4) *Montastrea annularis* forms very large heads, pillars and columns along the backreef edge of the lagoon and also on the deep forereef and bank; (5) *Porites astreoides* forms numerous small heads in the lagoon and numerous platelike

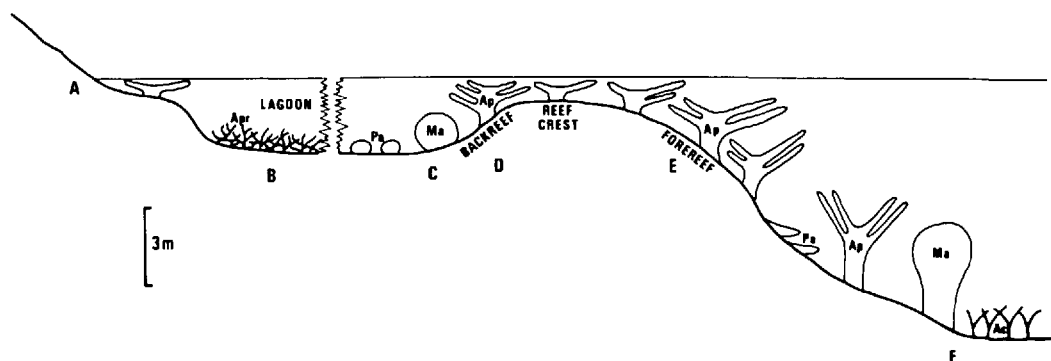


Figure 2. A schematic cross section of the eastern tip of Buck Island Reef showing the positions of the corals studied. Position: A. Buck Island, B. Lagoon, C. Backreef (2 m), D. Backreef ($\frac{1}{2}$ m), E. Shallow forereef and F. Deep forereef and bank. Corals: Ap *Acropora palmata*, Apr *A. prolifera*, Ac *A. cervicornis*, Ma *Montastrea annularis*, and Pa *Porites astreoides*.

colonies on the steeper portions of the fore-reef slope.

The corals were stained *in situ* with Alizarin Red S (Barnes, 1972, 1973). A clear plastic bag containing about 20 mg of the stain tied off in one corner was placed over the end of the branch (*Acropora* spp.) or over the entire colony. The bag was securely tied with cotton string. The final concentration of stain in the bag was 10–15 mg/l sea water (Dustan, 1975). The staining bags remained on the corals for 24 h except in the case of *A. palmata* where bags were removed after 4 h. Each stained coral was labelled with a plastic identifying tag attached to the sample with nylon monofilament. Corals were collected after 2–4 months, the tissue removed, and growth subsequent to staining determined as follows: (1) *Acropora* spp.: The linear extension of new skeleton was determined (by taking the mean of ten equally spaced measurements along the zone of new growth in *A. palmata*). Calcification rate of *A. palmata* was determined per branch perimeter by removing new (unstained) growth with a saw, weighing it, and dividing this by the perimeter of the branch at the base of the new growth. This value (g/cm/time) relates calcification to the original amount of calcifying tissue present; (2) Head corals:

Hemispherical corals were sectioned vertically through the center with a rock saw. The distance between the upper limit of the stain line and the periphery of the colony was measured using a dissecting microscope and ocular micrometer. The average of 10 such measurements spaced evenly across the top of the colony was used as a mean linear extension value for that colony. Gross surface area of each head was determined by the formula: $A = \frac{1}{2}\pi d^2$ where d = colony diameter. In the case of flattened colonies of *Porites astreoides* the surface area was calculated on the basis of a flat projected area.

RESULTS

The growth rates obtained in this study are summarized in Table 1, along with the principal environmental parameters considered. *Acropora palmata* is the only species which occurs abundantly in all three zones; *Montastrea annularis* and *Porites astreoides* both are common on the backreef and deep forereef. *Acropora prolifera* and *A. cervicornis* more or less replace one another in their respective zones in the lagoon and forereef-bank bottom. The only significant differences in linear extension rates as a function of an environmental parameter were seen in *Acropora palmata* and *A.*

Table 1. Growth rates and environmental conditions of five species of scleractinian corals from Buck Island, St. Croix, Virgin Islands

Reef Zone	Water Depth	Dates of Growth	N	Colony Size (ave.)	Linear Growth (mm/yr) ($\pm$ S.E.)	Water Temp. ($^{\circ}$ C)	Water Cond.	Light Calcification (rel. to g/cm/yr $\pm$ S.E. $\frac{1}{2}$ m)
<i>A. palmata</i>								
BR	$\frac{1}{2}$ m	8/5-9/30	12	large	58.5 ± 6.1	29.5	calm	1.0 0.82 ± 0.14
BR	$\frac{1}{2}$ m	2/10-4/5	14	large	55.4 ± 4.1	26	calm	1.0 0.87 ± 0.05
SFR	$\frac{1}{2}$ m	8/3-9/30	12	large	82.7 ± 6.9	29.5	surge	1.0 1.89 ± 0.20
SFR	$\frac{1}{2}$ m	1/20-4/5	8	large	47.3 ± 4.1	26	surge	1.0 1.32 ± 0.23
DFR	9m	8/3-9/30	12	large	99.3 ± 6.9	29.5	calm	0.3 1.39 ± 0.19
DFR	9m	1/20-4/5	25	large	65.4 ± 4.5	26	calm	0.3 1.33 ± 0.08
<i>A. prolifera</i>								
BR	2m	9/15-11/16	39		81.8 ± 3.2	29.5	calm	0.7
BR	2m	1/11-4/5	48		59.2 ± 2.2	26	calm	0.7
<i>A. cervicornis</i>								
DFR	10m	2/10-4/5	6		71.0 ± 6.5	26	calm	0.3
<i>M. annularis</i>								
BR	2m	8/12-11/10	16	—	$8.3 \pm .56$	29.5	calm	0.7
BR	2m	1/13-5/6	14	9.63 cm ²	$6.6 \pm .49$		calm	0.7
BR ave.					$7.6 \pm .33$			
DFR	10m	8/12-11/5	21	—	$7.0 \pm .42$	29.5	calm	0.3
DFR	10m	1/18-5/5	17	138 cm ²	$8.9 \pm .41$	26	calm	0.3
DRF ave.					$7.6 \pm .82$			
<i>P. astreoides</i>								
BR	2m	9/9-11/15	19	177 cm ²	$3.53 \pm .32$	29.5	calm	0.7
BR	2m	1/13-5/6	12	150 cm ²	$3.36 \pm .02$	26	calm	0.7
BR	2m	9/76-5/77	3	122 cm ²	$3.36 \pm .85$	26-29.5	calm	0.7
DFR	10m	1/26-5/5	4	95 cm ²	$3.00 \pm .12$	26	calm	0.3

prolifera. In the former, linear extension was significantly greater ($p < 0.05$) during the warm water period on the SFR and DFR sites. This was also true of the *A. prolifera* in the lagoon. *A. cervicornis* was not measured during the fall. During the warm water period only, linear extension of the branches of *A. palmata* was significantly greater on the SFR and the DFR than in the BR lagoon. Calcification rates of *A. palmata* were significantly higher ($p < 0.05$) on the SFR and the DFR than the BR during both the cool and warm water periods. By comparing calcification rate with linear extension rate in *A. palmata* (Table 1) it can be seen that the ratio of calcification to linear extension is greater during the cool water period in all three sites. Perhaps this

is due to greater density of skeletal material deposited at that time. Growth rates of *A. prolifera* and *P. astreoides* have not been reported heretofore. Rates in the other three species have been reported by other authors; a comparison of our rates with rates determined for other parts of the tropical western Atlantic are summarized in Table 2. The rates for *A. palmata* and *M. annularis* fall close to values obtained previously. The rates for *A. cervicornis* show greater variability.

DISCUSSION

Acropora palmata and *A. prolifera* show linear extension rates higher in the warm fall months than in the cooler spring months; Shinn (1966) reported a similar temperature

Table 2. A comparison of linear growth rates for corals of the western tropical Atlantic

Author	Technique	Location	Linear Growth (cm/yr)	Notes
<i>A. palmata</i> Present, 1977	Aliz. Red	Buck I., V.I.	5.5-5.8 4.7-8.7 6.5-9.9	BR ½m N = 26 SFR ½m N = 20 DFR 9m N = 37
Bak, 1976	Direct	Curacao	8.8 7.6	white tipped branches all branches N = 14
<i>A. cervicornis</i> Present, 1977	Aliz. Red	Buck I., V.I.	7.1	DFR 10m N = 6
Lewis et al., 1968	Direct	Barbados Jamaica	14.6 26.6	full year 10m N = 10 full year N = 10
Shinn, 1966	Direct	Florida	11	
Vaughn, 1915	Direct	Bahamas	4.5	
<i>M. annularis</i> Present, 1977	Aliz. Red	Buck I., V.I.	.76 .76	BR 2m N = 33 DFR 10m N = 38
Shinn, 1976	Direct	Florida	.84	spike
Bak, 1976	Direct	Curacao	.8	intact colonies
Dustan, 1975	Aliz. Red	Jamaica	.67	10m
Lewis et al., 1968	Direct	Barbados Jamaica	1.93 2.50	10m, transplanted fragments N = 10 transplanted fragments
Hoffmeister & Multer, 1964	Direct	Florida	1.07	3 yr study, 3m
Vaughn, 1916	Direct	Florida	.5-.68	N = 47
Hudson & Shinn (unpub.)	X-rad.	Buck I., V.I.	.73-.88	ave. of 12 cores, 18 yrs
Baker & Weber, 1975	X-rad.	St. Croix, V.I.	.92-1.04	3-7 m
Aller & Dodge, 1974	X-rad.	Jamaica	.62-.88	0-15 m
MacIntyre & Smith, 1974	X-rad.	Brit. Hon., Fla.	.66-.87	N = 3

related variation in the linear extension of *A. cervicornis* in which the rate of linear extension decreased 30-60% during those months in which the maximum water temperature did not exceed 26°C. His growth measurements for cold water months were very similar to those obtained in this study. Buddemeier and Kinzie (1975) found that *Porites lobata* in Hawaii also has the greatest linear growth in the warm summer months with a low density skeleton deposited at that time. Although skeletal density of *A. palmata* was not determined in this study our

results (by comparing rate of calcification to rate of linear extension) suggest that a denser skeleton may be deposited during the cold water months. Some investigators (MacIntyre and Smith, 1974) have suggested that in some corals (such as *Pavona* in Panama) high skeletal density might be related to warmer temperatures. Hudson et al. (1976) have conclusively shown that the high density bands of *Montastrea annularis* from Florida are deposited during the warmest months of the year. Thus, although warmer temperatures seem to promote a

greater rate of linear growth (in the branching corals considered), there is still no clear view of what factor or factors are most important in governing skeletal density.

With the exception of the studies by Lewis et al. (1968) most of the published linear extension values for *Montastrea annularis* are slightly less than 1 cm/yr. The anomalously high values obtained by Lewis may have been due to the fact that the studies were done on small transplanted fragments of colonies, whereas most of the other estimates are based on undisturbed larger heads. All the other rates for this species including those obtained in the present study are quite consistent regardless of geographical locality (Florida, Curaçao, Barbados, St. Croix, Jamaica), depth (down to 15 m where Baker and Weber, 1975, showed a steady decrease in growth increment from that depth downwards), local environment (shallow lagoon vs. forereef in this study and others) or method (Alizarin Red S, stakes, or X-radiography of sectioned heads). Hudson et al. (1976) have verified that periodic density variations in *M. annularis* are annual bands. The annual growth rates determined for *M. annularis* in the present study are very similar to rates determined from cores taken in the same area (Hudson and Shinn, unpub.).

The values obtained for *Acropora cervicornis*, on the other hand, vary greatly depending on geographical location, but probably more importantly on local environmental conditions. The value obtained in this study is very similar to that reported by Shinn (1966) when the time of measurement is considered. The values obtained by Lewis et al. (1968) are much higher. Growth in this species may be much more responsive to differing environmental conditions than *M. annularis*, with *A. palmata* falling somewhere in between. Previous growth rates have not been published for *A. prolifera* which is similar in morphology to *A. cervicornis* with more slender and more prolific branches. The linear extension rates for *A. prolifera* in the lagoon were

similar to those of *A. cervicornis* on the deeper bank.

Lewis et al. (1968) report rates of linear extension for *Porites porites* of 3.6 cm/yr, comparing this to rates of other branching *Porites* species of 2–4 cm/yr. *P. astreoides*, a mounding or platelike form, has a considerably lower linear extension rate of 3.0–3.4 mm of upward growth per year. It may be that growth is directed in a horizontal direction; this is suggested by the pictures of the same quadrats taken one year apart (Lewis, 1974) in which the colonies are seen to die off at one edge but advance in another direction.

It has been shown in the present study that for *Acropora palmata*, temperature as well as horizontal position on the reef (e.g. SFR vs. BR) are important in determining growth rate. In comparing our data with that of other investigators, species such as *Montastrea annularis* show relatively little response in growth rate to varying environmental conditions, whereas *Acropora cervicornis* shows much more variable rates. In conclusion, it seems that the growth of some coral species is sensitive to varying environmental conditions whereas other coral species are quite conservative in their rate of growth.

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